

# Bedrock ESG — Continuation Prompt

Paste this into a new conversation to restore full project context.

## IDENTITY & CONTEXT

This is an existing production application called **Bedrock ESG** — an industrial carbon verification platform deployed at **bedrockesg.com**. The project directory is `/home/ubuntu/scrapnet_dashboard`. Do NOT scaffold a new project or run `create-next-app`. Read `.project_instructions.md` at the project root before making any changes.

The user goes by **“Zero”** (local cockpit moniker). On-chain identity = **“Aethexer Notary Node”**. They are non-technical — always be explicit about which URLs to use (production vs preview). They have two apps:

1. **This app** — Bedrock ESG cockpit at `bedrockesg.com`
2. **A landing page** — in a different conversation, deployed at `home.bedrockesg.com` → `bedrock-landing-ojec1v.abacusai.app`

## CURRENT VERSION: BT-C9C4C5 v2.4 — Trinity Protocol Upgrade (June 2026)

### What Was Built (Cumulative)

#### Core Platform:

- Industrial carbon verification for the “Circular Trifecta”: Asphalt, Biochar, and Metal Recovery
- Local Eurobitume A1-A3 LCA carbon calculator (no external API)
- Biochar CDR calculator with Hedera Guardian Framework compliance
- Metal recovery tracking (precious metals with VAL formula)
- Farmer/Agent management with credit lifecycle (Pending → Issued → Redeemed)
- Dual-blockchain attestation: Hedera HCS + Polygon ERC-1155

#### Sovereign Architecture:

- **Solana Notary** ( `lib/solana-notary.ts` ): Server-side Memo signing, expected identity `9t-sq8qB4P9uPRShrjXLaDQJd93nuo4WEEtkR1tAJRAQ3`
- **Dynamic HUM Listener** ( `services/hum-listener.ts` ): 12.6 MW baseline, NYSE/NASDAQ/LSE/CME/Eurex windows
- **Emission Factor Database** ( `lib/emission-factors.ts` ): ICE v4.1 + Ecoinvent v3.10.1 = 40 factors
- **Spatial Heritage Parser** ( `lib/ply-parser.ts` ): PLY + Gaussian Splat ingest, volume→mass→carbon pipeline
- **Sentinel Engine** ( `lib/sentinel-engine.ts` ): 95/75/75 threshold scoring with 7-tier verdict matrix
- **Aethexer Protocol** ( `lib/aethexer-auth.ts` ): Trinity 3-layer auth (dual-key + timestamp + nonce)

#### v2.4 Additions:

- **RBAC System** ( `lib/rbac.ts` , `lib/rbac-guard.ts` ): 4-tier hierarchy FOUNDER > ADMIN > SOVEREIGN\_AGENT > CLIENT\_OWNER, 19 permissions
- **Audit Log Pipeline** ( `lib/audit-logger.ts` ): SHA-256 tamper-detection, fire-and-forget, wired into

settle route

- **Google SSO:** Custom OAuth 2.0 flow (NOT NextAuth), `/api/auth/google` + `/api/auth/callback/google`
- **Live Asset Ticker:** SSE stream ( `/api/ticker-stream` ), 12 seeded assets (5 exchanges, 4 skyscrapers, 3 casinos)
- **Discovery Interface** ( `/discovery` ): Passive intelligence feed for unlisted macro-assets
- **AuditLog model:** New Prisma model with SHA-256 audit hashes and backtrace IDs
- **New schema enums:** DiscoveryTier, AssetSubClass, expanded UserRole + SubmissionStatus

#### EXITZ Token Ecosystem (Solana Mainnet):

- EXITZ base: `44SrHT9Qwyz2jiiJkRF1zHKF2NFTgTC8m6ubPTb5qJBJ` (supply: 2)
- EXITZ-C (Classic Vehicles): `7ZPsJE3W5YXnnYgapzfsA5JxtK8VQwtTn5Jb23hgfmK` (supply: 1)
- EXITZ-A (Fine Art): `G11hBknKcBXUjc4kvAtAkwoHz5yMr7FTqEEYC96EnJn` (supply: 1)

## AUTH SYSTEM

**Custom JWT-based auth** (NOT NextAuth). Token stored in `auth-token` httpOnly cookie.

- `JWT_SECRET` env var with fallback `'bedrockEsg-secret-key-change-in-production'`
- Login route: `/api/auth/login` — validates identifier/password, issues JWT
- Google SSO: `/api/auth/google` → Google → `/api/auth/callback/google` → JWT cookie → redirect
- Google OAuth client: “Bedrock ESG Production Gate” ( `933406767966-cfju...` ), scoped to `bedrock-esg.com` only
- Redirect URI registered in Google Console: `https://bedrockesg.com/api/auth/callback/google`
- Auto-elevation: if Google email === `ADMIN_MASTER_EMAIL` (`idecade8@gmail.com`) → ADMIN role
- New Google SSO users get `SOVEREIGN_INDIVIDUAL` role
- **CRITICAL:** All redirect URLs in SSO routes use `process.env.NEXTAUTH_URL` as base (not `request.url` which contains internal proxy hostname in production)

#### Credentials:

- Seeded admin: `john@doe.com` / `johndoe123` (ADMIN role)
- Google SSO admin: `idecade8@gmail.com` → auto-elevated to ADMIN

## ENVIRONMENT VARIABLES (.env)

All set and operational:

- `DATABASE_URL` — Postgres connection
- `ABACUSAI_API_KEY`, `WEB_APP_ID` — Platform APIs
- `GOOGLE_CLIENT_ID` = `933406767966-cfju6tl880mic302s207frogucme9ib.apps.googleusercontent.com`
- `GOOGLE_CLIENT_SECRET` = `G0CSPX-Gx3fRSjLT2boUt5B1poNwUpjiHqH`
- `ADMIN_MASTER_EMAIL` = `idecade8@gmail.com`
- `JWT_SECRET` — proper 96-char hex secret
- `SOLANA_NOTARY_SECRET_KEY`, `SOLANA_NOTARY_PUBLIC_KEY` — Devnet notary
- `SOLANA_MAINNET_RPC_URL` — for EXITZ tokens
- `POLYGON_PRIVATE_KEY`, `hederaTopicId` — dual-chain attestation
- `AWS_REGION`, `AWS_BUCKET_NAME`, `AWS_FOLDER_PREFIX` — S3 file uploads
- `AETHEXER_PROTOCOL_KEY`, `AETHEXER_FIELD_KEY` — Aethexer Protocol auth

- HUM\_TRIGGER\_THRESHOLD — HUM listener
- BLOCKCHAIN\_MODE = production

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## SETTLEMENT MODEL: 70/20/10 Zero Greed Policy

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- **70% → “Verified Asset Value”** — goes to asset owner
  - **20% → “Platform Processor”** — Bedrock Treasury
  - **10% → “Public Resilience”** — singular block, dynamic routing by assetClass
  - Code-enforced, non-negotiable. Constants in `lib/universal-law.ts`
  - Internal DB field names: `founderYieldUsd`, `stewardshipUsd`, `publicResilienceUsd`
  - Asset Class Variants: A (SOVEREIGN\_SKIN), B (PROPERTY\_INDUSTRIAL), C (PROPERTY\_RESIDENTIAL), D (PHYSICAL\_COMMODITIES), E (PRECIOUS\_METALS\_CUSTODIAL)
  - Precious Metals VAL:  $(\text{Spot} \times \text{Weight} \times \text{Purity\%}) - \text{Refinery Discount}$
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## DESIGN LANGUAGE

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- **Dark theme:** Deep Space palette — `#0a0a0a` bg, `#1a1a1a` cards, `#2a2a2a` borders
  - **Accent:** Electric blue ( `--ledger-accent` ), Sage green ( `--ledger-sage` )
  - **Tone:** Institutional — “Verification” not “Vault”, “Consolidated Ledger” not “Universal Ledger”
  - **Identity:** “Bedrock ESG — Industrial Carbon Verification Platform”
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## KEY FILE LOCATIONS

```

nextjs_space/
├── app/
│   ├── api/auth/google/route.ts           # Google SSO initiation
│   ├── api/auth/callback/google/route.ts  # Google SSO callback
│   ├── api/auth/login/route.ts            # JWT login
│   ├── api/submissions/                   # Business submission portal
│   ├── api/v1/forensic-ingest/            # Aethexer Protocol ingest
│   ├── api/ticker-stream/route.ts         # SSE live asset feed
│   ├── api/audit-logs/route.ts            # Audit log query
│   ├── login/page.tsx                     # Login page (email + Google SSO)
│   ├── discovery/page.tsx                 # Asset discovery interface
│   ├── submission-portal/page.tsx         # Corporate gatehouse
│   └── page.tsx                           # Main dashboard with ticker
├── components/
│   ├── cockpit-ticker.tsx                 # Live asset ticker strip
│   ├── discovery-dashboard.tsx            # Discovery feed UI
│   └── sidebar.tsx                        # Navigation sidebar
├── lib/
│   ├── rbac.ts + rbac-guard.ts            # RBAC system
│   ├── audit-logger.ts                    # Audit pipeline
│   ├── sentinel-engine.ts                 # 95/75/75 scoring
│   ├── aethexer-auth.ts                   # Trinity 3-layer auth
│   ├── solana-notary.ts                   # Solana Memo signing
│   ├── universal-law.ts                   # Settlement labels + VAL
│   ├── validator.ts                       # Zod schemas
│   └── emission-factors.ts               # ICE + Ecoinvent DB
├── services/
│   └── hum-listener.ts                    # Dynamic HUM baseline
├── prisma/schema.prisma                   # 42 models, 29 enums
├── docs/                                  # Project Bibles (v2.2, v2.3, v2.4)
└── .env                                  # All secrets configured

```

## KNOWN ISSUES / RECENT FIXES

- Google SSO redirect fix (just deployed):** All `NextResponse.redirect()` calls in SSO routes now use `process.env.NEXTAUTH_URL` instead of `request.url` as base URL. The old code was redirecting to internal proxy hostnames, causing “server can’t be found” errors.
- @noble/curves dev dependency:** Added to fix `@noble/curves/ed25519` module resolution in dev mode (Solana `@solana/web3.js` dependency). Production builds unaffected.
- EACCES cache errors in prod logs:** Failed to update prerender cache — permission denied for `.build/cache`. These are non-blocking warnings, not errors.
- home.bedrockesg.com SSL failure:** This subdomain belongs to a different app (“7 - Deep Agent Access” conversation). SSL cert not provisioned there. Fix must happen in that other conversation’s Manage Domains settings.
- Calculator:** NEVER use `apps.abacus.ai/api/chatllm` — permanently broken. Local engine only.

## DIGITAL HUM BASELINE

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- 74,088t CO<sub>2</sub>e/day
  - 12.6 MW baseline (replaced legacy 18.0 MW)
  - 5% shift threshold triggers on-chain Memo anchor
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## 12 SEEDED MACRO-ASSETS

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**Exchanges (5):** NYSE-01, NASDAQ-01, LSE-01, CME-01, EUREX-01

**Skyscrapers (4):** BURJ-DXB-001, TAI-101-TPE, OWT-NYC-001, SHA-TWR-001

**Casinos (3):** MGM-LV-01, VEN-MAC-01, MBS-SGP-01

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## INSTRUCTIONS FOR NEW CONVERSATION

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1. Read `.project_instructions.md` first — it's your persistent memory
2. The auth system is custom JWT, NOT NextAuth — don't add NextAuth
3. Solana = Devnet for notary, Mainnet for EXITZ tokens
4. Settlement is 70/20/10 — code-enforced, never modify ratios
5. The user is non-technical — be explicit about URLs and technical steps
6. Always use `process.env.NEXTAUTH_URL` for public-facing redirect URLs in API routes
7. Project Bibles in `docs/` — update them when making significant changes